

I'll treat "destroy of cave" as **cave collapse / rockfall / ground movement / structural instability detection**. I'll separate practical sensor products from full monitoring systems, because caves usually need a system rather than one magic sensor.

Thought for 13s

Yes. There are sensor products for detecting **cave instability, tunnel deformation, rockfall, ground movement, and possible collapse**, but usually engineers use a **monitoring system with several sensor types**, not one sensor only.

For a cave, the useful sensor/product categories are:

Sensor / Product Type	What it detects	Example products / vendors
Convergence meter	Cave/tunnel walls moving closer together — strong sign of deformation	GEOKON Model 4425 Convergence Meter, used for underground excavations and tunnels
Extensometer	Internal rock/ground movement, crack opening, roof movement	GEOKON extensometers, RST multi-point borehole extensometers
Crack meter / displacement sensor	Crack widening or movement in 1D/3D	GEOKON Model 4415 3D Crackmeter
Inclinometer / tiltmeter	Rock wall, roof, or slope tilting before failure	GEOKON inclinometers / tiltmeters; RST tunnel profile monitoring system
LiDAR scanner / 3D laser monitoring	Shape change of cave wall/roof over time	Maptek Sentry uses 3D laser scan data to monitor rapid and gradual movement, including underground monitoring
Radar rockfall detection	Sudden falling rocks / moving rock mass	IDS GeoRadar RockSpot detects and tracks rockfalls
Fiber optic sensing cable	Distributed strain, vibration, ground movement along long distance	Solifos geotechnical monitoring cables; Sintela DFOS systems
Wireless geotechnical IoT logger	Sends sensor data remotely from cave/tunnel	Worldsensing geotechnical monitoring IoT system supports many sensor inputs and long-range communication

Seismic / acoustic sensor	Cracking, vibration, rock movement, microseismic activity	Often integrated with rockfall/landslide monitoring systems; INGLAS mentions inclinometers, seismometers, and extensometers together
Cave Tracker system	Mining cave propagation / ore movement inside cave mining	Elexon Cave Tracker monitors ore movement while still in the cave for near real-time cave behavior analysis

For your purpose, the best practical setup would be:

Basic low-cost monitoring

Use crack meters + tilt sensors + vibration sensors + datalogger. Good for detecting visible cracks, roof movement, and abnormal vibration.

Professional cave/tunnel safety monitoring

Use convergence meters + borehole extensometers + inclinometers + piezometers + wireless logger. This is the typical geotechnical engineering approach.

Advanced real-time warning

Use LiDAR or radar + fiber optic sensing + geotechnical sensors. This is better for high-risk caves, mines, railway tunnels, or tourist caves.

Important point: these sensors usually **do not “predict collapse perfectly.”** They detect warning signs: wall closure, roof sagging, crack growth, vibration, tilt, water-pressure change, and rockfall events. A geotechnical engineer normally sets alarm thresholds based on the cave geology.

For a cave, I would start with:

1. **Convergence meter** — detect cave/tunnel closure.
2. **Borehole extensometer** — detect roof/rock mass movement.
3. **Crack meter** — monitor cracks.
4. **Tiltmeter/inclinometer** — detect wall or roof rotation.
5. **Vibration/seismic sensor** — detect sudden rock movement.
6. **LiDAR scan every day/week** — compare 3D cave shape changes.

So yes, products exist — but the correct product depends on whether you want to monitor a **natural cave, mine cave, tunnel, tourist cave, or construction excavation.**

give me the download link for each item product catalog pdf

I'll look for **official catalog/datasheet PDFs first**, then use official product pages only where the manufacturer does not publish a direct PDF.

Thought for 13s

Yes — here are the **download links for catalog / datasheet PDFs** for each cave-collapse / tunnel-deformation monitoring item.

Item	Product / Vendor	PDF download link	Notes
Convergence meter	GEOKON Model 4425 VW Convergence Meter	Download PDF	For tunnel/cave closure deformation monitoring.
Borehole extensometer	GEOKON Model 1100 Rod-Type Borehole Extensometer	Download PDF	Measures internal rock/soil/concrete displacement.
Flexible borehole extensometer	GEOKON Model 1280 Flexible Rod-Type Borehole Extensometer	Download PDF	Useful where rigid rods are difficult to install.
Crack meter	GEOKON Model 4415 3D Crackmeter	Download PDF	Measures crack displacement in three directions.
In-place inclinometer	GEOKON Model 6150 MEMS In-Place Inclinometer	Download PDF	Detects ground/structure lateral deformation.
Digital inclinometer system	GEOKON GK-604D Digital Inclinometer System	Download PDF	Manual/profile measurement of borehole inclination.
Wireless tiltmeter	GEOSENSE / Worldsensing Loadsensing G7 Tiltmeter	Download PDF	Wireless 3-axis tiltmeter for slope, mining, and geotechnical monitoring.
Tunnel profile monitoring system	RST / Geosense Tunnel Profile Monitoring System	Download PDF	Linked rods fixed to tunnel wall to monitor deformation.
Tunnel concrete segment profile monitoring	RST Profile Monitoring System for Tunnel Concrete Segments	Download PDF	Uses tiltmeters on tunnel lining segments.

Tape extensometer	RST Tape Extensometer	Download PDF	Portable manual convergence/deformation measurement.
Wireless geotechnical data logger	Worldsensing Digital Logger	Download PDF	Connects digital geotechnical sensors to wireless monitoring network.
Fiber optic strain sensing cable	Solifos BRUsens DSS V9 Strain Sensing Cable	Download PDF	Distributed fiber optic strain monitoring cable.
Fiber optic strain sensing cable	Solifos BRUsens DSS 7.2mm V3 Grip	Download PDF	Another Solifos strain sensing cable option.
Cave mining movement tracking	Elexon CaveTracker System	Download PDF	For block cave mining / ore flow / cave propagation monitoring.
Cave propagation / subsurface monitoring	Elexon Geo4Sight	Download PDF	Wireless subsurface monitoring for caveback, propagation, and subsidence.
Full geotechnical catalog	GEOKON Instrumentation Catalog	Download PDF	Good general catalog covering convergence meters, extensometers, crackmeters, inclinometers, dataloggers, etc.
Full geotechnical catalog	RST Instruments Geotechnical Instruments Catalog	Download PDF	Older but useful RST catalog with many geotechnical instruments.

For **natural cave collapse detection**, the most relevant PDFs from this list are:

1. GEOKON 4425 Convergence Meter
2. GEOKON 1100 Borehole Extensometer
3. GEOKON 4415 3D Crackmeter
4. GEOKON 6150 In-Place Inclinometer
5. Tunnel Profile Monitoring System
6. Solifos fiber optic strain cable
7. Worldsensing wireless logger

For **mine cave / block cave monitoring**, the strongest match is **Elexon CaveTracker** and **Elexon Geo4Sight**.

give me the working principle for each item

I'll explain this in a practical engineering way: **what it measures, how it works, and what warning sign it gives**. I'll use the same product categories from the catalog list so it matches your table.

Thought for 4s

Below is the **working principle for each cave / tunnel / mine-collapse monitoring item** from the previous list.

1. GEOKON 4425 Convergence Meter

Purpose: Detects whether the cave/tunnel wall or roof is moving inward or outward.

Working principle:

Two anchor points are fixed on opposite sides of the cave, tunnel, or underground opening. A rod or wire connects the anchors to a displacement transducer. When the cave wall moves, the distance between the two anchors changes. The sensor converts that distance change into an electrical reading. GEOKON describes this as measuring contraction or elongation between two fixed anchor points in rock or soil masses.

What it tells you:

If the distance between walls becomes smaller, the cave may be squeezing or deforming. This can be an early warning sign before collapse.

2. GEOKON 1100 Rod-Type Borehole Extensometer

Purpose: Detects internal rock mass movement behind the visible cave wall or roof.

Working principle:

A borehole is drilled into the rock. One or more anchors are installed at different depths. Rods connect those anchors to a reference head near the surface. If deep rock layers move relative to the surface, the rod movement is measured by a mechanical dial, vibrating wire sensor, or electrical transducer.

What it tells you:

It shows whether the cave roof or surrounding rock is separating, settling, or deforming internally. This is more powerful than only watching surface cracks.

3. GEOKON 1280 Flexible Rod-Type Borehole Extensometer

Purpose: Same basic purpose as a rod extensometer, but easier to install in difficult or curved boreholes.

Working principle:

Flexible rods connect deep anchors to a measurement head. When the rock mass moves, the distance between anchor points and the reference head changes. The displacement is measured at the head.

What it tells you:

It helps detect internal roof movement, rock separation, or unstable zones where rigid rods are hard to install.

4. GEOKON 4415 3D Crackmeter

Purpose: Measures crack opening and movement in three directions.

Working principle:

Two halves of the instrument are mounted on opposite sides of a crack or joint. If the crack opens, closes, slides sideways, or moves vertically, the relative movement between the two sides is measured. GEOKON describes the 4415 as a rugged instrument for manually measuring displacement across cracks or expansion joints in three directions.

What it tells you:

If a cave crack is growing wider or shifting, the rock mass may be becoming unstable.

5. GEOKON 6150 MEMS In-Place Inclinometer

Purpose: Detects bending, lateral movement, or tilt inside the ground or cave wall.

Working principle:

A string of MEMS tilt sensors is installed inside an inclinometer casing or borehole. Each sensor measures the angle of one segment. By comparing tilt readings over time, the system calculates the deformation profile of the ground or rock. GEOKON says the system uses MEMS tilt sensors mounted on stainless-steel tubing and is designed for long-term deformation monitoring.

What it tells you:

It shows where the rock mass is bending or shearing. This is useful for detecting hidden sliding planes or unstable cave walls.

6. GEOKON GK-604D Digital Inclinator System

Purpose: Measures the shape/deformation profile of a borehole manually.

Working principle:

A probe is lowered into inclinometer casing at fixed depth intervals. At each interval, the probe measures tilt. The readings are combined to create a deformation profile along the borehole.

What it tells you:

It shows whether the ground or rock around the cave is slowly moving sideways. Unlike the in-place inclinometer, this is usually a manual periodic measurement tool.

7. GEONSENSE / Worldsensing Wireless Tiltmeter

Purpose: Detects tilt or rotation of rock surfaces, cave walls, tunnel linings, or support structures.

Working principle:

A MEMS accelerometer inside the sensor measures the direction of gravity. When the installed surface tilts, the measured gravity angle changes. The wireless logger sends the readings to a monitoring platform.

What it tells you:

If the cave wall, roof block, or support structure starts rotating, it may indicate instability or progressive failure.

8. RST / Geosense Tunnel Profile Monitoring System

Purpose: Monitors deformation of tunnel or cave lining/profile.

Working principle:

A series of connected rods or tilt sensors is installed along the tunnel lining or profile. When the tunnel/cave shape changes, the sensors detect angle or displacement changes. The software converts those changes into a profile deformation map.

What it tells you:

It shows whether the tunnel/cave section is deforming, squeezing, or losing its original shape.

9. RST Profile Monitoring System for Tunnel Concrete Segments

Purpose: Detects movement of tunnel concrete lining segments.

Working principle:

Tilt sensors are fixed to tunnel lining segments. If a segment rotates, shifts, or deforms, the sensor angle changes. Multiple sensors together show the deformation pattern of the tunnel lining.

What it tells you:

It helps detect lining distortion, joint movement, settlement, or structural deformation in a tunnel-like cave structure.

10. RST Tape Extensometer

Purpose: Portable/manual measurement of distance change between fixed points.

Working principle:

Reference anchors are installed on cave walls or tunnel surfaces. A precision measuring tape/extensometer is attached between two anchors. The operator measures the distance periodically and compares it with previous readings.

What it tells you:

If the measured distance changes over days/weeks/months, the cave opening is deforming. This is simple, cheap, and useful for routine inspections.

11. Worldsensing Digital Logger

Purpose: Collects sensor data and sends it remotely.

Working principle:

The logger connects to geotechnical sensors such as crackmeters, piezometers, tiltmeters,

extensometers, or inclinometers. It reads the sensor signal, stores the data, and transmits it wirelessly to a gateway or cloud platform.

What it tells you:

The logger itself does not detect collapse. It makes the monitoring system automatic, remote, and continuous.

12. Solifos BRUsens DSS V9 Fiber Optic Strain Sensing Cable

Purpose: Detects strain/deformation along a long cable.

Working principle:

A fiber optic cable is attached to rock, embedded in ground, or installed along a structure. A special interrogator sends light through the fiber. When the cable stretches, compresses, or bends, the light signal changes. The system calculates strain along the cable length.

What it tells you:

It can show where deformation is happening along a cave wall, tunnel, slope, or ground zone. This is useful when you need distributed monitoring, not just one sensor point.

13. Solifos BRUsens DSS 7.2 mm V3 Grip Fiber Optic Cable

Purpose: Similar to the DSS V9 cable, but with a different cable design for strain transfer and installation conditions.

Working principle:

The cable transfers ground or structure deformation into optical fiber strain. The interrogator reads strain distribution along the cable.

What it tells you:

It can detect long-zone deformation, settlement, rock movement, or strain concentration before visible failure.

14. Elexon CaveTracker System

Purpose: Tracks movement of broken ore/rock inside a block cave mine.

Working principle:

Magnetic beacons are embedded into the orebody. These beacons generate magnetic fields and can be tracked in 3D as they move with the fragmented rock. Elexon explains that the system uses embedded beacons that move with the rock mass, allowing engineers to determine which parts of the cave are moving.

What it tells you:

It shows cave flow behavior: which part of the cave is moving, where the ore/rock is flowing, and how the cave is propagating. This is mostly for **block cave mining**, not normal natural caves.

15. Elexon Geo4Sight

Purpose: Monitors cave propagation, caveback position, subsidence, and underground rock mass movement.

Working principle:

Geo4Sight is a wireless in-ground monitoring system. Sensors are installed in the ground/rock mass and report movement or damage-related data wirelessly. A technical paper describes Geo4Sight as a wireless in-ground system used for caveback and rock mass damage monitoring.

What it tells you:

It helps mine engineers understand where the cave is propagating, how the rock mass is changing, and whether ground movement is becoming dangerous.

Simple selection guide

Goal	Best sensor type
Detect cave wall closing	Convergence meter, tape extensometer
Detect crack growth	Crackmeter
Detect hidden roof/rock movement	Borehole extensometer
Detect wall/ground bending	In-place inclinometer

Detect surface rotation	Tiltmeter
Detect full tunnel/cave profile deformation	Tunnel profile monitoring system
Detect deformation over long distance	Fiber optic strain cable
Detect rockfall events	Radar, seismic/vibration sensor
Monitor mine cave propagation	CaveTracker, Geo4Sight
Send data remotely	Wireless logger

For a **natural cave**, I would not start with CaveTracker. I would start with:

crackmeter + convergence meter + borehole extensometer + tiltmeter + wireless logger.

For a **mine cave / block cave**, then **CaveTracker + Geo4Sight** becomes much more relevant.

what is the best for coal mine cave?

For a coal mine, the problem is a little different from a natural cave: the main danger is usually **roof fall, rib failure, goaf/caved-zone movement, floor heave, water pressure, and gas/ventilation risk**. I'll focus on sensors that are realistic underground, not fancy lab-only stuff.

Thought for 5s

For a **coal mine cave / underground coal mine**, the best monitoring choice is usually **not CaveTracker**. The best practical system is:

Best overall setup for coal mine collapse warning

1. Roof-to-floor convergence / closure sensor

This is the most direct sensor for coal mine entry stability. It measures whether the **roof and floor are moving closer together**. In coal mines, roof sag and floor heave are very important warning signs. NIOSH's ground-control guide specifically describes roof-to-floor extensometers for measuring convergence during active mining and long-term entry stability studies.

Best for:

Roof fall warning, entry deformation, longwall gate roads, room-and-pillar entries.

Example products:

GEOKON convergence meter, roof-to-floor extensometer, RST convergence/tape extensometer.

2. Multi-point borehole extensometer / roof extensometer

This is probably the **most important sensor for roof failure**. It is installed into the mine roof and measures movement at different depths. In coal mines, the visible roof may look okay, but separation can happen above the roof line. A borehole extensometer can detect that hidden separation earlier. NIOSH roof-monitoring literature notes that roof and rib extensometers are widely used for detecting separations and roof-beam deflection.

Best for:

Roof separation, weak strata movement, roof sag, bolted roof performance.

Example products:

GEOKON rod extensometer, RST borehole extensometer, tell-tale roof extensometers.

3. Rib extensometer / rib displacement sensor

Coal mine “walls” are usually called **ribs**. Rib failure can be very dangerous, especially in room-and-pillar retreat mining. A coal rib monitoring study used extensometers in both the roof and ribs to measure vertical roof sag and horizontal rib movement.

Best for:

Rib spalling, pillar/rib deformation, retreat mining danger zones.

4. Microseismic monitoring system

For larger coal mines, this is the best advanced warning system. It detects tiny cracking events inside the coal/rock mass before large failure. A 2024 review says microseismic monitoring is widely used in coal mines for mine-safety management, including assessment of mining-induced instability.

Best for:

Rockburst risk, roof instability, goaf/caved-zone movement, large-area monitoring.

Important:

Microseismic is powerful, but it needs expert interpretation. It is not a simple plug-and-play alarm.

5. Stress meter / instrumented rock bolt

This measures load change in roof bolts or surrounding rock. If bolt load increases quickly, the roof is taking more stress and may be unstable. GEOKON lists instrumented rockbolts for underground mines to measure loads in supports used to stabilize ground.

Best for:

Checking whether roof support is overloaded.

My recommendation by priority

Priority	Sensor	Why it is important in coal mine
1	Multi-point roof extensometer	Detects roof separation before visible collapse
2	Roof-to-floor convergence sensor	Detects entry closure from roof sag/floor heave
3	Rib extensometer	Detects coal rib/pillar side failure
4	Instrumented rock bolt / stress meter	Shows roof support loading
5	Microseismic system	Best for large-area early warning and goaf/rockburst behavior
6	Wireless data logger + alarm system	Makes the system real-time and automatic

Best simple system

For a normal underground coal mine section, I would choose:

Roof extensometer + roof-to-floor convergence meter + rib extensometer + wireless logger + alarm threshold software.

That is the practical “core system.”

Best advanced system

For longwall mining, deep mining, retreat mining, or high-risk roof conditions:

Roof extensometers + convergence sensors + instrumented rock bolts + microseismic monitoring + gas/water monitoring + real-time control room dashboard.

Not my first choice for coal mine

CaveTracker / Geo4Sight are more for **block cave metal mining**, not typical underground coal mining. They track cave propagation or ore flow, which is not usually the main safety-monitoring method for coal mine entries.

So, the best answer is:

For coal mine cave/roof collapse detection, start with multi-point roof extensometers and roof-to-floor convergence sensors. Add microseismic monitoring if the mine is large, deep, longwall, or high-risk.

give me the catalog pdf for these products

I'll gather **catalog/datasheet PDFs for the coal-mine-relevant sensors only**: roof extensometers, convergence meters, rib displacement/extensometers, instrumented rock bolts/stress meters, microseismic systems, and wireless loggers.

Thought for 14s

Here are the **catalog / datasheet PDF links** for the coal-mine collapse / roof-fall monitoring products I recommended.

Coal mine monitoring need	Product / vendor	PDF catalog / datasheet
Roof separation / roof movement	GEOKON Model 1100 Rod-Type Borehole Extensometer	Download PDF
Roof / rib / tunnel deformation	Encardio Rite EDS-70V Multi-Point Borehole Extensometer	Download PDF
Roof / rock mass displacement	Geosense GEO-XB2 Borehole Rod Extensometer	Download PDF
Roof-to-floor convergence / mine entry closure	GEOKON Model 4425 Convergence Meter	Download PDF
Manual convergence measurement	RST Tape Extensometer	Download PDF
Rock bolt load / support stress	GEOKON Model 4910 Instrumented Rockbolt	Download PDF
Rock bolt / cable bolt load monitoring	Mine Design Technologies SMART Cable Bolt	Download PDF

Rock bolt / cable bolt installation guide	MDT SMART Cable Bolt Installation Instructions	Download PDF
Tilt / rotation of roof, rib, support, or structure	GEOSENSE Tiltmeter	Download PDF
Remote sensor data collection	Worldsensing Digital Logger	Download PDF
Remote analog sensor data collection	Worldsensing Analog Logger	Download PDF
Wireless vibrating-wire sensor logger	Worldsensing Vibrating Wire Logger	Download PDF
General geotechnical / mine instrumentation catalog	GEOKON Instrumentation Catalog	Download PDF
Coal-mine ground-control monitoring guide	NIOSH Ground Control Monitoring Guide	Download PDF
Coal mine microseismic monitoring reference	Microseismic Monitoring of Underground Coal Mines	Download PDF
Mine seismic monitoring guide	Institute of Mine Seismology — Guide to Seismic Monitoring in Mines	Download PDF
Coal mine microseismic application example	ESG Solutions — Applications of Microseismic Monitoring in China's Underground Coal Mines	Download PDF

My strongest recommendation for a **coal mine roof/cave collapse warning system** is to start with these PDFs first:

1. **GEOKON 1100 Borehole Extensometer** — roof separation.
2. **GEOKON 4425 Convergence Meter** — roof-to-floor closure.
3. **GEOKON 4910 Instrumented Rockbolt** — support load.
4. **Worldsensing Digital / Vibrating Wire Logger** — automatic remote readings.
5. **NIOSH Ground Control Monitoring Guide** — practical coal mine monitoring layout.